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Programming In C-HDOC DAY_14

Problem

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Check whether all its digits are unique.
 - **E.g. 1:** Number: 12345, Output: Unique
 - **E.g. 2:** Number: 12234, Output: Not Unique
- Find the frequency of each digit from `0` to `9` .
 - **E.g.** Number: 112333, freq_0 = 0, freq_1 = 2, freq_2 = 1, freq_3 = 3, freq_4 = 0, freq_5 = 0, freq_6 = 0, freq_7 = 0, freq_8 = 0, freq_9 = 0

ALGORITHM

: Start.

Step 2: Declare an integer variable c and a character array num[100].

Step 3: Ask the user to enter a positive integer number.

Step 4: Read the number as a string and store it in num.

Step 5: Display the two available choices:

- Choice 1: Check whether all digits are unique.
- Choice 2: Find the frequency of each digit from 0 to 9.

Step 6: Read the user's choice into c.

Step 7: Use a switch statement based on c.

If c = 1

Step 8: Initialize index = 0.

Step 9: Compare each digit with the digit immediately following it.

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Step 10: If two consecutive digits are equal, increment index.

Step 11: After all comparisons:

- If index == 0, display "**unique**".
- Otherwise, display "**not unique**".

Step 12: Exit this case.

If c = 2

Step 13: Set i = 0 and repeat until i < 10.

Step 14: Initialize count = 0 for the current digit.

Step 15: Compare every digit in num with the current digit i.

Step 16: If num[j] == i + '0', increment count.

Step 17: Display the frequency of digit i.

Step 18: Increment i and repeat Steps 14–17 for digits 0 through 9.

Step 19: Exit this case.

If the choice is invalid

Step 20: If c is neither 1 nor 2, display "**invalid choice**".

Step 21: Stop.

C PROGRAM CODE

```
#include <stdio.h>
```

```
#include <math.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    int c;
```

```
    char num[100];
```

```
    printf("enter a positive integer number\n");
```

```
    scanf("%s", num);
```

```
    printf("choice1 is to find whether all digits of the numbers are unique\n");
```

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```
printf("choice 2 is to find the freq of each digit from '0 to 9'\n");
printf("enter CHOICE number\n");
scanf("%d", &c);
switch (c)
{
    case 1:
    {
        int index=0;
        for(int i=0;i<strlen(num)-1;i++)
        {
            if(num[i]==num[i+1])
            {
                index++;
            }
        }
        if(index==0)
            printf("unique\n");
        else
            printf("not unique\n");
        break;
    }
    case 2:
    {
        for(int i=0;i<10;i++)
        {
            int count=0;
            for(int j=0;j<strlen(num);j++)
```

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```
{
    if(num[j]==i+'0')
        count++;
}
printf("frequency of %d is %d\n", i, count);
}
break;
}
default:
printf("invalid choice\n");
break;

}
return 0;
}
```

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TEST CASES

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	enter a positive integer number 111111 choice1 is to find whether all digits of the numbers are unique choice 2 is to find the freq of each digit from '0 to 9' enter CHOICE number 1	not unique	not unique	PASS
2	enter a positive integer number 112233454 choice1 is to find whether all digits of the numbers are unique choice 2 is to find the freq of each digit from '0 to 9' enter CHOICE number 2	frequency of 0 is 0 frequency of 1 is 2 frequency of 2 is 2 frequency of 3 is 2 frequency of 4 is 2 frequency of 5 is 1 frequency of 6 is 0 frequency of 7 is 0 frequency of 8 is 0 frequency of 9 is 0	frequency of 0 is 0 frequency of 1 is 2 frequency of 2 is 2 frequency of 3 is 2 frequency of 4 is 2 frequency of 5 is 1 frequency of 6 is 0 frequency of 7 is 0 frequency of 8 is 0 frequency of 9 is 0	PASS